

A Simulation Study on Airfoils Using VAWT Design for Low Wind Speed Application

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Abstract— This paper presents the simulation study on the airfoil based on the vertical axis wind turbine (VAWT) for low wind speed application ranging 2 to 8 meter per seconds. The main advantage VAWT compared to the horizontal axis wind turbine (HAWT) is the ability to capture wind from any direction. Therefore, the yaw system is not required to turn the rotor towards the wind direction. Furthermore, the noise level produced during the rotation making it suitable to be installed on residential and urban environment. However, the major drawbacks of VAWT over HAWT are less efficient due to drag and turbulent force that the blade produces and low starting torque. The aim of this study is to develop a VAWT design equipped with the advantages mentioned and to overcome the VAWT drawbacks. The National Advisory Committee of Aeronautics (NACA) for both airfoil series 0012 and 0015 were analyzed using simulation provided in the Qblade software. The graphical relation between optimal angles of attack (AoA) and the Reynolds number which predicts the flow patterns of the wind is analyzed. Based on this, the AoA that generate high lift over drag ratio (Cl/Cd) can be identified. The ratio of Cl/Cd for the airfoils; NACA 0012 and NACA 0015 are used to analyze the relationship between Tip Speed Ratio (TSR) and the Power Coefficient (Cp) performance. The number of blades is also tested to justify the effect on the power produced in low wind speed condition. Comparison of the simulated airfoils is presented in this paper. A simulated design of small scale size VAWT is made to test the effect of the blades number and their respective Cp and TSR versus wind speed. With the proper airfoils design that suits the low wind speed condition, the turbine can functions efficiently as well as maximize the power produced by turbine.

Keywords- VAWT, HAWT, NACA series, AoA, Reynolds number, TSR, Cl/Cd ratio and Cp

I. INTRODUCTION

In a few decades, the non-renewable energy sources such as coal, gas, and oil will be limited and the production will be minimized [1]. Due to high in demand of these sources, there will be an increment in price and gives great effect to well

develop countries. Non-renewable energy has some global climate change issue and carbon dioxide as well as sulphur dioxide emission from its production [2]. The renewable energy generation is the solution that is being implemented to replace non-renewable energy as primary sources of power generation [2]. Renewable energy is cost effective, low to non-carbon emission, climate friendly and unlimited. Some forms of renewable energy are solar, tidal, biomass and bio-fuel, geothermal, and wind. Wind turbines are widely used all over the world and it is regarded as an alternative source for electrical power generation [3]. Wind turbine energy return of energy invested (EROEI) and can be considered high among the renewable energy generation with value of 8, compared to solar power which is 5 [3].

Wind turbine can be classified into two types; horizontal axis wind turbine (HAWT) and vertical axis wind turbine (VAWT). VAWT is the new discovery and has the ability to capture wind from any directions. Savonius is the VAWT type and incorporated the drag force. Small wind turbines are generally those rated at 10kW or less. Wind turbines system consists of a turbine, a tower, a controller, a grid connected inverter and a meter. Malaysia falls on the region in Southeast Asia which has a very low wind speed which is below 5m/s [4].

Problem of wind turbine is associated with the designing of the wind turbine itself, some of the wind turbine are designed is too big to harness more energy from the wind. This is true if the wind speed is high but for country with a low wind speed, designing a small wind turbine is more efficient in harnessing the wind [5, 6]. Some of the researchers have started to harness the power from wind by designing a small wind turbine which can operate at low wind speed with low Reynolds number [7, 9]. However, it might be impossible to even make the rotor spin and harness great energy, due to the low wind speed.

II. AIRFOIL AND WIND TURBINE THEORY

In a wind turbine design, airfoil configuration for blades is one of the most important factors for high efficiency of the turbine to convert kinetic energy from wind into mechanical

energy using rotor and generator that converts mechanical energy to electrical energy. The cut in speed is when the rotor starts to generate power, while the cut out speed is when the rotor start to reduce its speeds and power generated [8, 9]. The details are as shown in Figure 1.

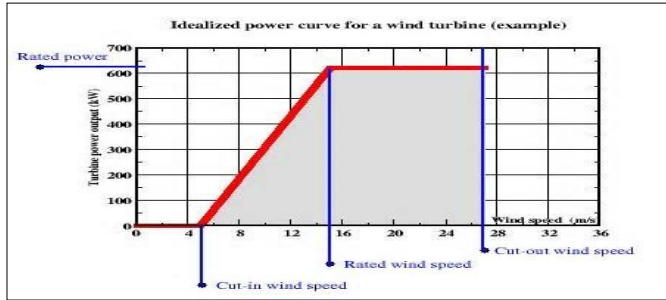


Fig.1. Relationship of wind turbine against wind speed

The 'Lift coefficient' (Cl) is the lifting force generated by the airfoil and 'Drag coefficient' (Cd) is the drag or resistance force generated in the airfoil. These two forces are related to each other and important in giving precise information on the lift to drag ratio (Cl/Cd) in the airfoil configuration. Both forces are commonly associated with the 'Angle of Attack' (AoA) to determine the forces to be high or low at a particular AoA. The maximum Cl for a certain AoA for an airfoil is called stall angle. The common stall angle for airfoils is between 10 to 15 degrees [10].

Tip Speed Ratio (TSR) related to the number of blades used in the wind turbine. The number of blades used influence the performance of the turbine. This is due to the effects on the solidity of the wind turbine. The equation for rotor solidity is shown as follows [7]:

$$\sigma = \frac{NbC}{r} \quad (1)$$

Where σ is solidity, Nb is number of blades, C is the length of blade chord and r is radius of the blade to rotor.

The number of blades varies from 3, 5 and 8 blades and the performance in terms of the power produced is also studied. With higher number of the blades, the TSR value is reduced. Thus increasing the drag force, less power is produced and practically less efficient. The relationship between TSR and the number of blades used can be deduced using the formula shown [7]:

$$TSR_{\max} = \frac{4\pi}{Nb} \quad (2)$$

III. SIMULATION SET UP

A simulation was done to compare the performance between the two airfoils using QBlade software. The airfoils employs the National Advisory Committee on Airfoils (NACA) four digit series; NACA 0012 and NACA 0015. These airfoils are selected because due to the performance of the Cl/Cd ratio although both have low starting torque. The airfoils configuration is shown in Table 1 and illustrated in Figure 2.

TABLE I
THICKNESS OF NACA 0012 AND NACA 0015

Airfoil Designation	Thickness	Camber
0012	12.00	0
0015	15.00	0

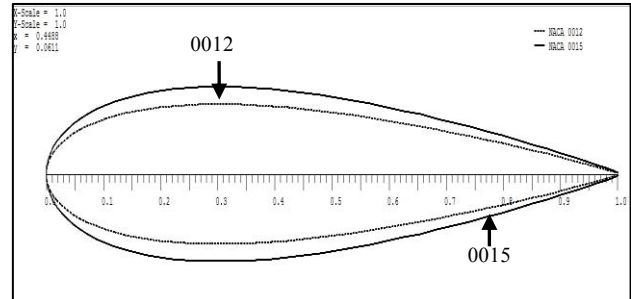


Fig. 2. The comparison of shape of NACA 0012 and NACA0015

The analysis comprises of the lift (Cl), drag (Cd) and lift to drag ratio (Cl/Cd) of the airfoils against a set range AoA in set of Reynolds number (Re) ranging from 50000 up to 1000000.

A rotor-blade was designed and simulated using the data from the airfoils to test the power coefficient (Cp) against TSR. The simulation was done using three blades with different airfoils. The specification for the rotor-blade is based on data in Table 2 and the design is based on Figure 3.

TABLE 2
ROTOR-BLADE SPECIFICATION

Height of the turbine	0.66m
Chord of the blade	0.167m
Radius of the blade from the rotor	1.00m

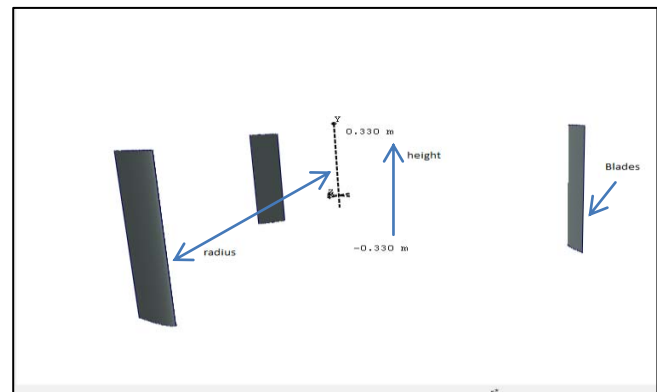


Fig. 3 Side view of the rotor blade design

IV. RESULTS AND DISCUSSION

The simulation result shows that NACA 0015 has better performance in terms of Cl/Cd ratio compared to the NACA 0012. The result of optimal AoA for the both airfoils is clearly shown in Table 3.

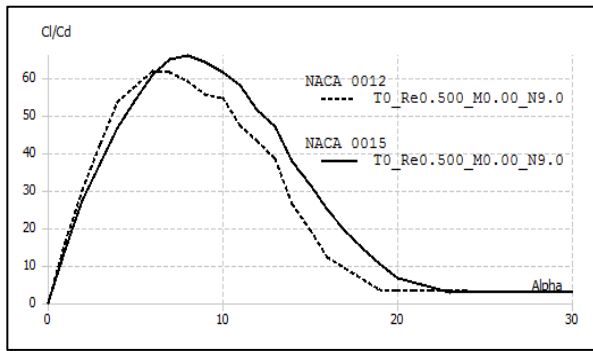


Fig. 5 Comparison of Cl/Cd versus AoA for NACA 0012 and NACA 0015 in 50000 Re

TABLE 3

OPTIMAL ANGLE OF ATTACK IN RELATION TO LIFT/DRAG RATIO IN 500000 RE

Reynolds number	Airfoil	Max Cl/Cd
50000	NACA 0012	25.7 at $\alpha=5^\circ$
50000	NACA 0015	24.7 at $\alpha=6.25^\circ$
100000	NACA 0012	36.7 at $\alpha=5^\circ$
100000	NACA 0015	37.5 at $\alpha=6^\circ$
200000	NACA 0012	47.4 at $\alpha=5^\circ$
200000	NACA 0015	49.6 at $\alpha=6.25^\circ$
500000	NACA 0012	61.7 at $\alpha=6.5^\circ$
500000	NACA 0015	66.4 at $\alpha=7.5^\circ$
1000000	NACA 0012	75.6 at $\alpha=7.5^\circ$
1000000	NACA 0015	77.9 at $\alpha=9^\circ$

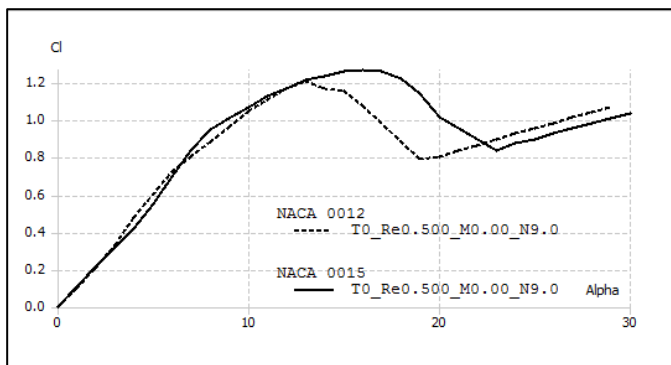


Fig.6. Comparison between Cl against AoA for both airfoils

Table 3 shows the optimal angle of attack for max Cl/Cd for both airfoils at Re range from 50000 to 1000000. The results shows that NACA 0015 have better Cl/Cd from Re range from 50000 to 100000 except 50000 in which NACA 0012 performance are slightly better. Cl/Cd is important for the airfoils to stay elevated and not dragged by the force. In wind turbine, it is important as it keep the blade spin.

Figure 6 show the comparison between Cl against AoA for both airfoils in Re of 50000. The Cl for NACA 0015 is better than NACA 0012 with Cl_{max} approximately higher than 1.2 at AoA of around 15 degrees. NACA 0012 produces lower Cl_{max} which is below 1.2 at AoA around 12 degrees. Cl is the ability of the airfoils to lift when exposed to any force act upon it. For Darrieus, it is more preferable to have a high Cl value.

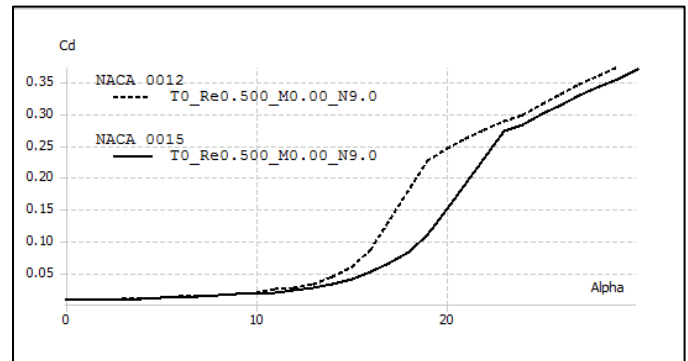


Fig. 7 Comparison between Cd against AoA for both airfoils.

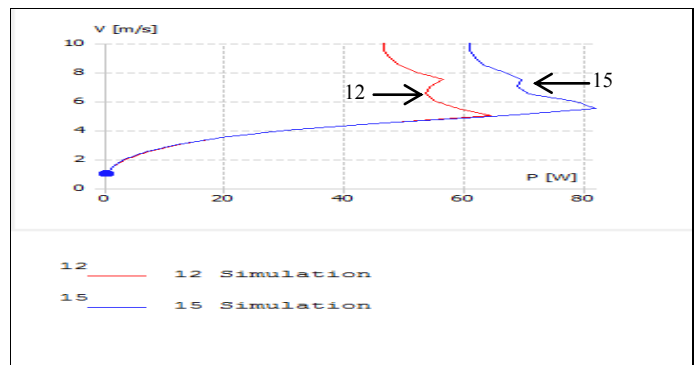


Fig. 8. Comparison of power generated against wind speed for NACA 0012 and NACA 0015 with three blades and rotor solidity of 0.51.

The value of Drag coefficient (Cd) in Figure 7 of NACA 0012 was higher than NACA 0015, in most cases of wind turbine design, the drag forces is not desirable, except for drag type turbine such as Savonnius.

The difference in performance lies when producing output power. In Figure 8, NACA 0015 produced better output power in higher wind speed, the difference between both airfoils is the thickness in which NACA 0012 has a thickness of 12 compared to NACA 0015 which has a higher thickness of 15. The thickness of the airfoil affects the performance of the rotor in a higher wind speed.

The performance of the rotor against the number of blades was simulated. The variation of three, five and eight blades were simulated and compared for both NACA 0012 and NACA 0015 airfoils. The results was shown in the Figures 9 to Figure 12. The NACA 0015 has better performance compared to NACA 0012 and was selected to explore the effects of the number of blades against power coefficient.

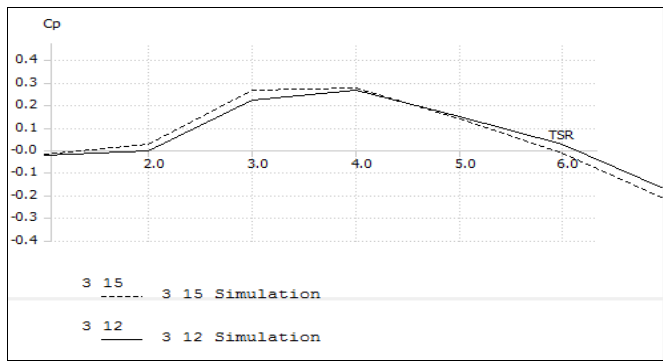


Fig. 9. Comparison of Cp against TSR for 3 blades

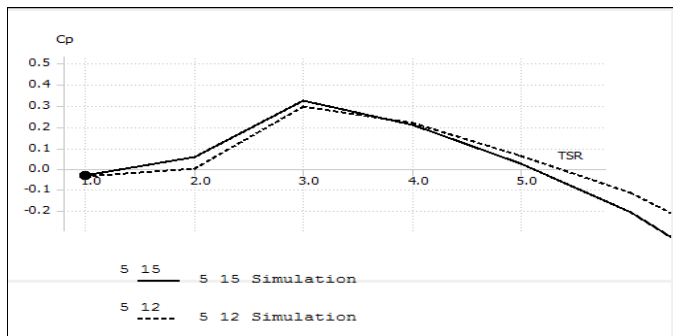


Fig. 10 Comparison of Cp against TSR for 5 blades

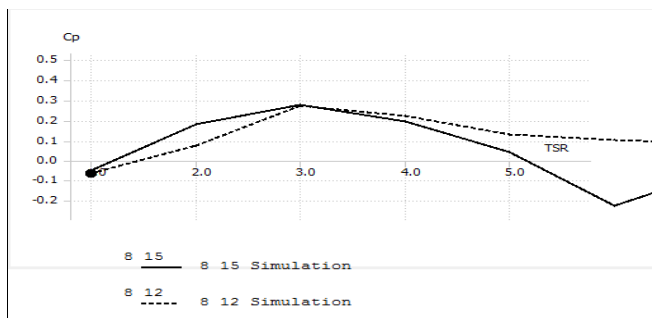


Fig. 11 Comparison of Cp against TSR for 8 blades

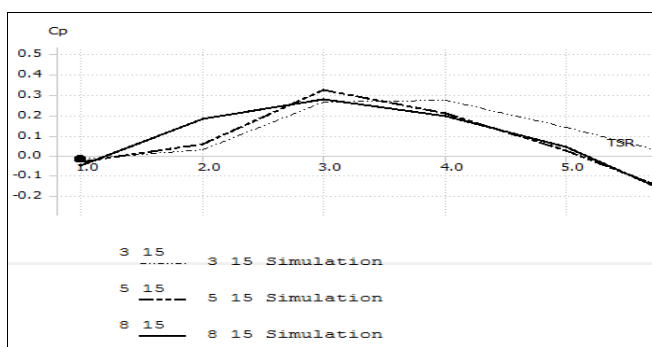


Fig. 12 Comparison of Cp against TSR of NACA 0015 for 3, 5 and 8 blades.

The Cp versus TSR curves varied with different number of blades used. Three blades produces a higher and much stable power

coefficient if compared to higher number of blades used. The value of maximum power coefficient for three blades is about 0.3 at TSR within range 3 to 4. The five and eight blades have the maximum value of Cp at around 0.3 and 0.2 respectively. The TSR has the maximum value occurred at 3 for 5 and 8 blades, the only difference is the Cp for eight blades is higher at TSR value of 2. The TSR_{max} can also be calculated using a simple equation in Section 2. Table 4 shown the value of TSR_{max} using calculation.

TABLE 4
CALCULATION VALUE FOR TSR_{max} FOR VARIOUS NUMBER OF BLADES

Number of blades	TSR_{max}
3	4.2
5	2.5
8	1.6

The solidity of the rotor also affects the performance of the turbine, using the solidity equation in 1, the solidity value can be determine for the numbers of blades selected. For the three, five and eight blades, the solidity values are 0.51, 0.85 and 1.36 respectively. Based on this, the higher number of blades gives high solidity value which causes higher drag force. Three blades give an optimal solidity which can produce more power due to high performance of Cp.

The torque of the rotor against different wind speed was also investigated for the number of blades used. It was found that the higher the number of blades results in better torque in low wind speed condition, torque is important for the turbine to be self-rotating and work as stand alone application. The drawback of high number of blades is the their efficiency to produce power. Low number of blades has high efficiency and power coefficient but due to inability of low number of blades to self rotate, it is advisable to hybrid it with drag type rotor to overcome the self-starting problem.

The higher number of blades produces more drag force than the lower number of blades. Furthermore, the value of TSR is low but high in torque in low wind speed condition which means it is self-operated at low wind speed condition. But it has low power efficiency compared to lower number of blades. If the TSR is low, it is hard to match with the generator which operates at low TSR.

V. CONCLUSION

The conclusion that can be made from the the simulation study is:

- NACA0015 show a better performance under a high wind speed compared to the NACA0012. The optimal stall angle is around 10 to 15 degrees for both airfoils. The power output produce for NACA 0015 is higher than NACA 0012 under higher wind speed.
- The number of blades affects the TSR and Cp produced by the wind turbine. Three blades are optimal and have stable power coefficient than five and eight. Higher number of blades employed more drag forces and torque, making it self starting and efficient under low wind speed condition.

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